

# MATH 102 - Applied Linear Algebra

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## 1 Invertible Matrix Theorem

### 1.1 Definition

Let  $A$  be a square  $n \times n$  matrix. Then the following statements are equivalent. That is, for a given  $A$ , the statements are either all true or all false.

1.  $A$  is invertible.

2.  $A$  is row equivalent to  $I$ .
3.  $A$  has  $n$  pivot positions.
4. The equation  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution.
5. The columns of  $A$  are linearly independent.
6. The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  is one-to-one.
7. The equation  $A\mathbf{x} = \mathbf{b}$  has at least one solution for each  $\mathbf{b}$  in  $\mathbb{R}^n$ .
8. The columns of  $A$  span  $\mathbb{R}^n$ .
9. The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  maps  $\mathbb{R}^n$  onto  $\mathbb{R}^n$ .
10. There is an  $n \times n$  matrix  $C$  such that  $CA = I$ .
11. There is an  $n \times n$  matrix  $D$  such that  $AD = I$ .
12.  $A^T$  is invertible.
13. The columns of  $A$  form a basis of  $\mathbb{R}^n$ .
14.  $\text{Col } A = \mathbb{R}^n$ .
15.  $\dim \text{Col } A = n$ .
16.  $\text{rank } A = n$ .
17.  $\text{Nul } A = \{0\}$ .
18.  $\dim \text{Nul } A = 0$ .
19. The number 0 is *not* an eigenvalue of  $A$ .
20.  $\det A \neq 0$ .
21.  $(\text{Col } A)^\perp = \{0\}$ .
22.  $(\text{Nul } A)^\perp = \mathbb{R}^n$ .
23.  $\text{Row } A = \mathbb{R}^n$ .

## 1.2 Thm 9.1.16

Let  $A \in M_n$  and let  $\lambda \in \mathbb{C}$ . Then the following statements are equivalent:

- (a)  $\lambda$  is an eigenvalue of  $A$ .
- (b)  $A\mathbf{x} = \lambda\mathbf{x}$  for some nonzero  $\mathbf{x} \in \mathbb{C}^n$ .
- (c)  $(A - \lambda I)\mathbf{x} = \mathbf{0}$  has a nontrivial solution, that is,  $\text{nullity}(A - \lambda I) > 0$ .
- (d)  $\text{rank}(A - \lambda I) < n$ .
- (e)  $A - \lambda I$  is not invertible.
- (f)  $A^\top - \lambda I$  is not invertible.
- (g)  $\lambda$  is an eigenvalue of  $A^\top$ .

### Proof

- (a)  $\Leftrightarrow$  (b)

By definition 9.1.1, if  $A\mathbf{x} = \lambda\mathbf{x}$  and  $\mathbf{x} \neq \mathbf{0}$  then  $(\lambda, \mathbf{x})$  is an eigenpair of  $A$ , meaning  $\lambda$  is an eigenvalue of  $A$ .

- (b)  $\Leftrightarrow$  (c)

These are restatements of one another.

- (c)  $\Leftrightarrow$  (d)

By rank nullity theorem, if  $\text{nullity}(A - \lambda I) > 0$ , then  $\text{rank}(A - \lambda I) < n$ .

- (d)  $\Leftrightarrow$  (e)

By property 1 and 16 of IMT, if  $\text{rank}(A - \lambda I) < n$  then  $(A - \lambda I)$  is not invertible.

- (e)  $\Leftrightarrow$  (f)

Suppose we have  $A - \lambda I = \begin{bmatrix} a - \lambda I & b \\ c & d - \lambda I \end{bmatrix}$ , and suppose it is not invertible, its determinant is  $(a - \lambda I)(d - \lambda I) - bc = 0$ . Consider the case of  $A^\top - \lambda I$ , we have  $\begin{bmatrix} a - \lambda I & c \\ b & d - \lambda I \end{bmatrix}$ , whose determinant is  $(a - \lambda I)(d - \lambda I) - cb$ . And by commutativity of multiplication,  $bc = cb$ , so the determinant is also 0, meaning  $A^\top - \lambda I$  is also not invertible.

(f)  $\Leftrightarrow$  (g)

Since (a) is equivalent to (e), the same is true for  $A^\top$ .

## 2 Annihilator

### 2.1 Theorem 9.3.3

Let  $A \in \mathbb{M}_n$  and let  $p$  be a nonconstant polynomial that annihilates  $A$ . Every eigenvalue of  $A$  is a root of  $p(z) = 0$ .

**Proof:**

Let  $(\lambda, \mathbf{x})$  be an eigenpair of  $A$ . Then  $\mathbf{0} = 0_n \mathbf{x} = p(A)\mathbf{x} = p(\lambda)\mathbf{x}$ , so  $p(\lambda)\mathbf{x} = 0$

#### 2.1.1 Is The Converse True?

While all the eigenvalues of a matrix are roots of each of its annihilators, the roots of an annihilator are not necessarily eigenvalues.

This raises a question:

Must some of the roots of an annihilator of a matrix be eigenvalues of the matrix? The answer is yes.

Let  $p(z) = a_0 + a_1z + a_2z^2 + \dots + a_kz^k$  be an annihilator of a matrix  $A \in \mathbb{M}_n$ . Factor  $p$  as follows:  $p(z) = a_k(z - \lambda_1)(z - \lambda_2) \dots (z - \lambda_k)$ ,  $a_k \neq 0$   
Since  $p$  annihilates  $A$ , we have

$$0 = p(A) = a_k(A - \lambda_1 I)(A - \lambda_2 I) \dots (A - \lambda_k I)$$

Since the zero matrix is not invertible, one of the factors  $A - \lambda_i I$  is not invertible. This ensures that  $\lambda_i$  is an eigenvalue of  $A$ .

## 2.2 Each Matrix has an Eigenvector

### 2.2.1 Proof 1

$\lambda$  is an eigenvalue of  $A \in \mathbb{M}_n$  iff  $\det(A - \lambda I) = 0$ .  
 $p(\lambda) = \det(A - \lambda I)$  is a polynomial of degree  $n$  (the characteristic polynomial

of  $A$ ).

Therefore,  $\lambda$  is an eigenvalue of  $A$  iff  $\lambda$  is a root of  $p$  (i.e.  $p$ )

By the Fundamental Theorem of Algebra (FTA), each polynomial over the complex has a root.

Hence, each matrix  $A \in \mathbb{M}_n$  has an eigenvalue.

### 2.2.2 Proof 2

We've shown that each matrix  $A \in \mathbb{M}_n$  has an annihilator  $p$ .

By the FTA,  $p$  must have a root.

In 2.1, we showed that at least one root of  $p$  is an eigenvalue of  $A$ .

Hence, each matrix  $A \in \mathbb{M}_n$  has an eigenvalue.

## 3 Eigenvalues

### 3.1 Lemma 9.3.2

Let  $(\lambda, x)$  be an eigenpair of  $A \in \mathbb{M}_n$  and let  $p$  be a polynomial. Then  $p(A)x = p(\lambda)x$ , that is,  $(p(\lambda), x)$  is an eigenpair of  $p(A)$ .

**Proof:**

Recall  $A^k \mathbf{x} = \lambda^k \mathbf{x}$ .

$$\begin{aligned} p(A)\mathbf{x} &= (c_k A^k + c_{k-1} A^{k-1} + \dots + c_1 A + c_0 I)\mathbf{x} \\ &= c_k A^k \mathbf{x} + c_{k-1} A^{k-1} \mathbf{x} + \dots + c_1 A \mathbf{x} + c_0 I \mathbf{x} \\ &= c_k \lambda^k \mathbf{x} + c_{k-1} \lambda^{k-1} \mathbf{x} + \dots + c_1 \lambda \mathbf{x} + c_0 \mathbf{x} \\ &= (c_k \lambda^k + c_{k-1} \lambda^{k-1} + \dots + c_1 \lambda + c_0) \mathbf{x} \\ &= p(\lambda) \mathbf{x} \end{aligned}$$

### 3.2 2-Commute

Let  $A_1, A_2 \in \mathbb{M}_n$ . Let  $\lambda$  be an eigenvalue of  $A_1$ . Suppose  $A_1 A_2 = A_2 A_1$ , then  $A_1, A_2$  share an eigenvector in  $\mathcal{E}_\lambda(A_1)$

**Computation:**

0. Choose  $A_1$  or  $A_2$ . Say,  $A_1$

1. Compute an eigenvalue of  $A_1$ . Say,  $\lambda$ , and form its eigenspace  $\mathcal{E}_\lambda(A_1)$
2. Select a basis  $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$  for  $\mathcal{E}_\lambda(A_1)$  and form a matrix  $X = [\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_k]$
3. Solve  $A_2 X = X C$  for  $C$
4. Find an eigenpair of  $C : (\mu, u)$
5. Compute  $v = Xu$

**Proof:**

$$Ax_1 = \lambda x_1, \quad Ax_2 = \lambda x_2, \quad \dots, \quad Ax_k = \lambda x_k$$

$$A\vec{x}_i = \lambda \vec{x}_i \quad \text{for } i = 1, 2, \dots, k$$

$$A_1 x_i = \lambda x_i$$

$$A_2(A_1 x_i) = A_2(\lambda x_i)$$

$$A_1(A_2 x_i) = \lambda(A_2 x_i) \quad (\text{commutativity})$$

Case 1:  $A_2 x_i = 0$   
 $\Rightarrow A_2 x_i \in \mathcal{E}_\lambda(A_1)$

Case 2:  $A_2 x_i \neq 0$   
 $\Rightarrow A_2 x_i$  is an eigenvector of  $A_1$  and  $A_2 x_i \in \mathcal{E}_\lambda(A_1)$

$$A_2 x_1 = X c_1, \quad A_2 x_2 = X c_2, \quad \dots, \quad A_2 x_k = X c_k$$

$$A_2[x_1 \ x_2 \ \dots \ x_k] = [A_2 x_1 \ A_2 x_2 \ \dots \ A_2 x_k] = [X c_1 \ X c_2 \ \dots \ X c_k] = X[c_1 \ c_2 \ \dots \ c_k]$$

$$A_2 X = X C$$

Let  $(\mu, u)$  be an eigenpair of  $C$   $C\vec{u} = \mu\vec{u}$   $u \neq 0$

Say  $v = Xu \neq 0$

$$Cu = \mu u$$

$$XCu = X\mu u$$

$$A_2 Xu = \mu Xu$$

$$A_2 v = \mu v$$

$\therefore v = Xu \in \mathcal{E}_\lambda(A_1)$  is an eigenvector of  $A_2$

### 3.3 Gershgorin Disk Theorem

If  $n \geq 2$  and  $A = [a_{ij}] \in \mathbb{M}_n$  then

$$\text{spec } A \subseteq \mathcal{G}(A) = \bigcup_{k=1}^n \mathcal{G}_k(A).$$

**Proof:**

Let  $(\lambda, \mathbf{x})$  be an eigenpair of  $A = [a_{ij}] \in \mathbb{M}_n$  and let  $\mathbf{x} = [x_i]$ . Since  $\lambda \mathbf{x} = A\mathbf{x}$ , we have

$$\begin{aligned} \lambda x_i &= \sum_{j=1}^n a_{ij} x_j = a_{ii} x_i + \sum_{j \neq i} a_{ij} x_j \\ (\lambda - a_{ii}) x_i &= \sum_{j \neq i} a_{ij} x_j \end{aligned}$$

Let  $k \in \{1, 2, \dots, n\}$  be any index such that  $|x_k| = \max_{1 \leq i \leq n} |x_i|$ . Since  $\mathbf{x} \neq \mathbf{0}$ , it follows that  $|x_k| > 0$  and  $\frac{|x_j|}{|x_k|} \leq 1$  for all  $j = 1, 2, \dots, n$ . Set  $i = k$  and divide by  $x_k$  to obtain

$$\lambda - a_{kk} = \sum_{j \neq k} a_{kj} \frac{x_j}{x_k}$$

By the triangle inequality,

$$|\lambda - a_{kk}| = \left| \sum_{j \neq k} a_{kj} \frac{x_j}{x_k} \right| \leq \sum_{j \neq k} |a_{kj}| \left| \frac{x_j}{x_k} \right| \leq \sum_{j \neq k} |a_{kj}| = R'_k(A)$$

Therefore,  $\lambda \in \mathcal{G}_k(A) \subseteq \mathcal{G}(A)$

### 3.4 Corollary 9.4.15

Let  $A \in \mathbb{M}_n$ . If  $A$  is strictly diagonally dominant, then it is invertible.

**Proof:**

The strict diagonal dominance of  $A$  ensures that  $0 \notin \mathcal{G}_k(A)$  for each  $k = 1, 2, \dots, n$ . The Gershgorin Disk Theorem then tells us that 0 is not an eigenvalue of  $A$  and therefore  $A$  is invertible.